

Module 4: A+ Troubleshooting and Helpdesk

DAY 3:-

Task 1:

Five Common BIOS/POST Error Messages

Error Message	What it indicates
"CMOS checksum error - Defaults loaded"	Failing/dead CMOS battery or BIOS settings became corrupted.
"No bootable device found"	Storage drive not detected, wrong boot order, or corrupted/missing OS
"Overclocking failed" / "CPU over temperature error"	Unstable overclock settings or inadequate CPU cooling
"Keyboard error or no keyboard present"	Keyboard not connected properly, faulty port, or (on older systems) missing keyboard entirely
"BIOS ROM checksum error"	Corrupted BIOS firmware, often from a failed update or power loss during flashing

Task 2:

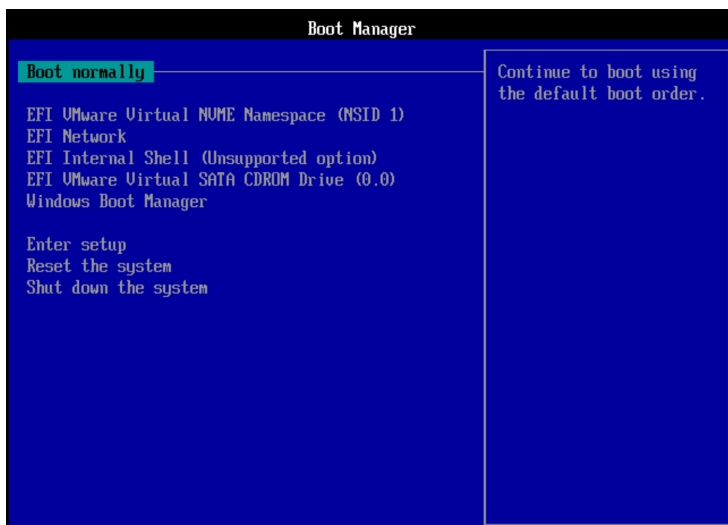
3 Beeps, No Display (AMI BIOS)

Three short beeps on AMI BIOS indicate a base memory read/write test error in the first 64KB block of memory — this is commonly summarized as a "Base 64K RAM failure." In short: this is a RAM problem.

Two troubleshooting steps:

1. Power off, reseal the RAM module firmly in their slots, and try booting again — a loose connection is the most common cause.
2. If reseating doesn't help, try replacing the RAM module — test with a single known-good stick if you have access to spares, to isolate whether one specific module is faulty.

Task 3:



BIOS Version

BIOS Version/Date

VMware, Inc. VMW201.00V.24006586.BA64.2406042154, 04-06-2024

Motherboard Manufacturer

BaseBoard Manufacturer

VMware, Inc.

Task 4:

"CMOS Checksum Error – Defaults Loaded"

What it means:

This error indicates the BIOS detected that its stored settings are **corrupted or lost**, and has automatically reverted to factory default settings to allow the system to boot. This is almost always caused by the **CMOS battery**.

How to resolve it in a real system:

1. **Check the system date/time** — if it's reset to a default/incorrect date (a classic symptom of a dead CMOS battery), that confirms the battery is the culprit.
2. **Replace the CMOS battery** — power off and unplug the PC, locate the battery near the PCIe/SATA area of the motherboard, and swap it for a fresh one in the same orientation.

3. **Re-enter any custom BIOS settings** — since defaults were loaded, you'll need to reconfigure anything you'd customized before.
4. **If the error persists after replacing the battery** — the issue may instead be a fault in the CMOS chip itself or a deeper motherboard issue, which would require further diagnosis or professional repair.